Topology-aware learning for higher-order cell-cell interaction in spatial transcriptomics: Introduction to spatial transcriptomics

Lucia Testa, Robin Khatri

Institute of Medical Systems Bioinformatics, UKE

LOGML summer school

This module

Bringing a machine-learning and mathematics background to a live problem in spatial biology.

- **Part 1 (this talk).** What spatial transcriptomics is, the data, the libraries, the standard methods, and where they break.
- **Part 2 (labs).** Two notebooks: an introduction to the data, libraries and workflow, then spatial methods and the open challenges.
- **Part 3.** The higher-order / topological project (next part).

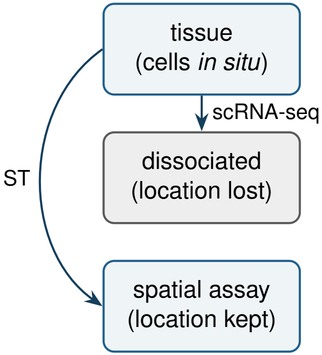
Repo: `https://github.com/robinredX/st-topo-aware-learning-logml`

Why spatial?

Dissociation-based scRNA-seq tells you *which* cells are present but throws away *where* they were.

Tissue function lives in **organisation**: who sits next to whom, tissue compartments, local signalling hubs. Spatial transcriptomics (ST) keeps the coordinates, so we can ask those questions directly.

For an ML audience: ST hands you a **geometry** (points in a plane) with a high-dimensional signal (expression) attached, plus rich structure to exploit.



Two families of technology

Spot / **capture based** (Visium, Slide-seq, HD)

- barcoded spots on a slide; *whole transcriptome*
- a spot mixes several cells (Visium $\sim$1–10); needs deconvolution
- coarse spatial resolution

Imaging / **probe based** (Xenium, MERFISH, CosMx)

- image individual transcripts; *single-cell, subcellular*
- a fixed gene **panel** ($\sim$100–500 genes)
- needs cell **segmentation**

The trade-off: **transcriptome-wide but low resolution** vs **single-cell but targeted**. This module uses **Xenium** (single-cell, so cells are real nodes).

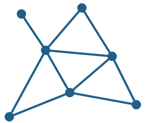
Marx. Method of the Year 2020. *Nat Methods* 2021; Moses & Pachter. Museum of spatial transcriptomics. *Nat Methods* 2022.

The data object

Everything is an **AnnData**: a cell $\times$ gene matrix plus annotations.

- `adata.X` – expression (cells $\times$ genes)
- `adata.obs` – per-cell table (cell type, sample, QC)
- `adata.obsm['spatial']` – x, y coordinates
- `adata.obsp[...]` – the **spatial graph**

The spatial graph (Delaunay / kNN / radius) turns coordinates into a discrete structure every downstream method builds on.



cells $\rightarrow$ nodes, adjacency $\rightarrow$ edges

The libraries

- **scanpy** / **anndata** – the single-cell backbone (IO, QC, normalisation, clustering, embeddings).
- **squidpy** – spatial extension: neighbour graphs, neighbourhood enrichment, co-occurrence, Ripley, `ligrec` for communication.
- **spatialdata** / **spatialdata-io** – readers and a coordinate-aware container for Xenium/Visium (images, shapes, points).
- **torch + torch-geometric** – graph neural nets on the spatial graph.
- **gudhi** / **toponetx** / **topomodelx** – topology: simplicial complexes, persistence, and higher-order message passing (Part 2).
- **liana** / **decoupler** – ligand-receptor resources and consensus CCC.

Wolf et al. Scanpy. *Genome Biol* 2018; Palla et al. Squidpy. *Nat Methods* 2022.

The core tasks

cell typing → spatial domains → niches (recurring units) → cell-cell communication

- **Spatial domains**: partition the tissue into coherent regions from expression *and* position.
- **Niches**: recurring multi-cell neighbourhoods (the composition is the label).
- **Communication**: which cell types signal to which, and where.

These are the tasks a topology-aware model can target.

Spatial domains: the method landscape

Almost all are **graph** methods, expression smoothed or attended over the spatial graph:

- **SpaGCN** – graph convolution over spots + histology.
- **STAGATE** – graph attention autoencoder.
- **BANKSY** – augment each cell with its neighbourhood mean and azimuthal gradient, then cluster (fast, strong baseline).
- **CellCharter** / **nichePCA** – cluster neighbourhood-aware embeddings into niches.

Common thread: a node feature plus message passing on the spatial graph, the graph falls out naturally from the coordinates.

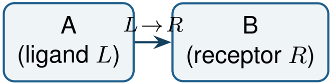
Hu et al. SpaGCN. *Nat Methods* 2021; Dong & Zhang. STAGATE. *Nat Commun* 2022; Singhal et al. BANKSY. *Nat Genet* 2024; Varrone et al. CellCharter. *Nat Genet* 2024.

Cell-cell communication: the standard logic

Score a **ligand-receptor (LR)** pair: ligand L high in cell type A, receptor R high in a neighbouring type B.

- **Non-spatial:** CellPhoneDB, CellChat, NicheNet; LIANA unifies them.
- **Spatial-aware:** squidpy `ligrec`, stLearn, COMMOT (optimal transport), NCEM.

The score lives on an **edge** $A - B$. It is **pairwise by construction.**



Efremova et al. CellPhoneDB. *Nat Protoc* 2020; Jin et al. CellChat. *Nat Commun* 2021; Dimitrov et al. LIANA. *Nat Commun* 2022; Cang & Nie. COMMOT. *Nat Methods* 2023.

Challenges (where you come in)

- **Segmentation.** Cell boundaries are estimated; mis-assigned transcripts corrupt co-expression and communication.
- **Sparsity and noise.** Few counts per cell/gene; targeted panels miss genes.
- **Scale.** Millions of cells per slide; methods must be graph-sparse and batched.
- **Batch / patient effects** across samples and conditions.
- **Beyond pairwise.** Graphs and LR edges capture pairwise relations; relays and multi-cell niches are higher order, the focus of the project (next part).

Where this module goes: beyond pairwise

The pairwise picture misses interactions that are intrinsically **higher order**:

- **Relay signalling.** $A \to B \to C$: the meaningful unit is the chain, which a pairwise scan splits into two unrelated hits.
- **Multi-cell niches.** A glomerular crescent or an immune focus is a coordinated group of three or more cell types, not a single edge.

$$\boxed{\mathrm{A}} \xrightarrow{L_1 \to R_1} \boxed{\mathrm{B}} \xrightarrow{L_2 \to R_2} \boxed{\mathrm{C}}$$

This is the project's focus: **CellNEST** (attention-based relay networks) and **topological deep learning** (message passing over triangles and higher cells). **Lucia covers both in detail next.**

The dataset

Human kidney Xenium (GSE294965): ANCA vasculitis, lupus nephritis, anti-GBM and control biopsies; ∼3.2M cells, single-cell resolution.

- One AnnData (`GSE294965_processed_data.h5ad`, ∼3.9 GB). Download once, point `ST_DATA` at it (see `data/datasets.md`).
- Disease context makes higher-order structure biologically real: crescents, immune infiltrates, injured tubules are multi-cell units.
- Prefer a smaller tissue? Any 10x Xenium public dataset works (`10xgenomics.com/datasets`). The notebooks only need coordinates + a cell-type label.

Ligand-receptor resources ship in `data/`: a gene-symbol LR table + the canonical CellPhoneDB.

The notebooks

- **01 – introduction.** Load a Xenium section, the AnnData object, QC, normalisation, cell types, spatial visualisation.
- **02 – spatial methods and challenges.** Neighbourhood graph, neighbourhood enrichment, domains (nichePCA), a brief `ligrec`, and the open challenges.

The higher-order / topological project builds on this in the next part. `src/topo_utils.py` ships the shared helpers (spatial graph, graph-to-complex lift) for it.

Project directions

- **Higher-order CCC.** Lift LR scoring to triangles; do high-scoring motifs localise to known niches (crescents, immune foci)?
- **Topology vs graph.** Same section, same task (domain segmentation): GCN/GAT baseline vs a TopoModelX simplicial model. Where does higher order help?
- **Denoising / reconstruction.** Mask genes or cells; does higher-order propagation recover signal better than pairwise?
- **Benchmark the inductive bias.** When is the extra machinery worth it?

References

- Marx. Method of the Year 2020: spatially resolved transcriptomics. *Nat Methods* 2021; Moses & Pachter. Museum of spatial transcriptomics. *Nat Methods* 2022.
- Palla et al. Squidpy. *Nat Methods* 2022; Wolf et al. Scanpy. *Genome Biol* 2018.
- Hu et al. SpaGCN 2021; Dong & Zhang. STAGATE 2022; Singhal et al. BANKSY 2024; Varrone et al. CellCharter 2024.
- Efremova et al. CellPhoneDB. *Nat Protoc* 2020; Jin et al. CellChat 2021; Dimitrov et al. LIANA. *Nat Commun* 2022; Cang & Nie. COMMOT. *Nat Methods* 2023.
- Fatema et al. CellNEST. *Nat Methods* 2025.
- Bodnar et al. WL go cellular / topological. *ICML/NeurIPS* 2021; Papillon et al. TopoModelX / TopoNetX 2023; Hajij et al. Topological deep learning 2023.